Replication of 'An Adaptive Perspective on Visual Working Memory Distortions' by Chunharas, Rademaker, Brady, and Serences (2022, Journal of Experimental Psychology)

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Introduction

I am largely interested in the study of memory from a cognitive perspective, as there is still much to explore about the features and boundaries of working memory in all forms. I have previously focused on much more rudimentary motor skill learning tasks, which have introduced

me to creating my own web experiment with JavaScript, HTML, and CSS. With the paper I chose being focused on visual working memory (VWM), I feel sufficiently challenged by coding a completely different kind of paradigm in a format I'm partially familiar with. The first experiment in the paper yielded a very strong confirmation of repulsion as a VWM phenomenon, as the results showed participants were "better at rejecting a foil color that was distorted toward the non-target memory item than rejecting a foil color that was distorted away from the non-target memory item" (Chunharas, 2022).

For this replication, I will need to reproduce the 2-AFC color wheel VWM task from Experiment 1 in an online format, able to be conducted on SONA. The task involves a continuous color wheel, where I will have to display 3 color samples from the wheel: one target color, and two adjacent colors 6^{o} in color space away from the target. The participant will then be asked to choose the correct target from two colors, one of which will be a foil. For this task, I will have to re-code it in Javascript, HTML, and CSS, since the original experiment used MATLAB and Psychophysics toolbox. Thankfully, the first experiment was also conducted online, so the format will be perceptually indistinguishable for the participant. I can foresee many hurdles when reproducing this task in a different language, however by utilizing the "Stimuli & Procedure" section under Experiment 1, I will be able to make sure the task is identical in presentation. The post-hoc data analysis I will be conducting will be a paired t-tests on accuracy. The original results showed toward vs. away gave $t(44)=3.98$, $p<.001$, toward vs. chance gave $t(44)=3.73$, $p=.006$, and away vs. chance gave $t(44)=-1.76$, $p=.08$.

The first task I coded was much simpler than this, and was concerned mainly with precision of recording input, not precision with displaying stimuli, so working with changing stimuli on a screen will be new. In my original experiment, I also received a substantial amount of help with the code structure, debugging, and most of the work in HTML and CSS.

I hope to improve my coding skills, and refresh my knowledge on these 4 languages I will be using. While this task is slightly different from my existing research, I wanted to use this new task to develop a work flow from start to finish. I would like to learn the proper pipeline for carrying out a project, including preregistering, coding organization, data collection, data formatting, analysis, and paper drafting.

Here is a link to the data from the paper: https://osf.io/qp6xk/files/r2pgq?view_only=0559769c587c4c8294288451

Link to Replication Repository: https://github.com/jamesrashby21/chunharas2022

Link to Replication Paper: https://dx.doi.org/10.1037/xlm0001014

Methods

Link to original code: https://osf.io/qp6xk/?view_only=0559769c587c4c8294288451e8af239e

Task will be coded in Javascript, HTML, and CSS, adapted to the SONA online platform. Data analysis and visualization will be done in R.

Power Analysis

Effect size is not displayed in the original paper, but can be extrapolated from data and figures later. Power analysis needed to confirm viability of sample size.

Planned Sample

The sample will consist of UC San Diego undergraduates, recruited via the SONA platform and compensated with class credit for their participation. The anticipated sample will be pre-screened to ensure they have normal or corrected-to-normal vision without color-blindness. The estimated size of the sample will be 50 people, which is only slightly more than the original population. The larger sample population is due to extraneous noise that can sometimes occur during online experiments among a student population.

Materials

All materials consist of the code for the task and the online SONA platform. There are exact directions on how to display the visual stimuli for the experiment.

Procedure

Experimental procedure pulled from Chunharas, 2022: "Participants were shown two memory items for 150ms at two randomly selected placeholders in the display (out of 12 possible placeholders), with the restriction that there were always at least 2 empty placeholders between the two memory items. After a 750ms delay, a location cue (arrow) indicated which of the items was the memory target, and two response options appeared in the placeholders directly to the left and right of fixation. One of the response options was always the correct color (i.e. identical to the color that was cued), while the other option was always a foil, and participants made a 2-AFC judgment between the two response options. The foil always differed from the correct color by 6o in color space, either in the direction towards (50% of trials) or away (50% of trials) from the non-target memory item. The positions (left or right of fixation) of the correct and foil response options were completely randomized. There were 60 trials per condition (a total of 120 trials per participant)."

Analysis Plan

Data analysis will consist of simply running a paired samples t-test comparing accuracy between the two foil conditions in the 2-AFC task.

I also plan to visualize the data myself, and see if I can plot new figures if reaction time was recorded.

```
setwd("~/yourWorkingDirectory/.../data_codes/R") # <- please change this to your own path

#### EXP1 ####
dat = 'EXP1_subj_fdist_fdirec_val.csv'
datcsv = read.csv(dat, header = FALSE)
colnames(datcsv) = c('subj','fdist','fdirec','val')

datcsv$subj = as.factor(datcsv$subj)
datcsv$fdirec = as.factor(datcsv$fdirec)

# t-test
t.test(datcsv[datcsv$fdirec==-1,'val']-datcsv[datcsv$fdirec==1,'val'])
```

Differences from Original Study

There will be slight variations to this sample, in that the original sample consisted of individuals recruited from Amazon Mechanical Turk. We will differ in that this will likely be a younger, WEIRD-er, university student sample. Since we are talking about fundamental processing systems that constitute visual memory, I do not expect any differences in the results or conclusions, but I will need to ensure proper pre-screening to rule out color-blind participants.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.